

AI-NATIVE SPORTS INTELLIGENCE PLATFORM

CourtVision

The System Bible — every moving part, how it works, how it gets better, what it honestly is today, and where it can go.

Computer Vision → Possession Sim

4 Sports · One Engine Each

18-Daemon Self-Healing Stack

Self-Improving · Leak-Free Gates

Obsidian Brain · Auto-Memory

Honest by Construction

Architect & Director: **Neel Shah** — solo human directing an agentic build pipeline.

Built Mar–May 2026 · ~1,470 commits · neeljshah22@gmail.com

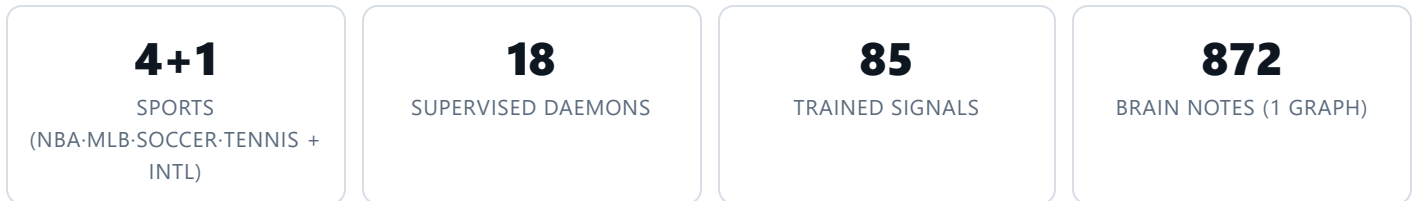
Calibration, not profit. Markets are efficient; we match the close. No dollar edge is claimed in this document.

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1 • Executive Summary

CourtVision is an AI-native NBA intelligence platform evolving into a domain-agnostic, multi-sport forecasting and decision engine. It turns raw sport data — including its own computer-vision tracking — into **calibrated probabilistic predictions**, validates every step against a brutal leak-free gate, and improves itself within hard honesty rails. It runs unattended, heals itself, and tells the truth about what it does and does not know.



The defining idea is the **funnel**: every data point is pushed through as many candidate signals as possible, the survivors feed models, the models feed one engine per sport, the engine emits one coherent prediction per market, and an intelligence layer plus an agentic loop re-validate the whole chain continuously. The funnel was originated here — building this kind of end-to-end, self-validating sports-detail pipeline is genuinely hard precisely because it has not been done this way before.

The honest north star

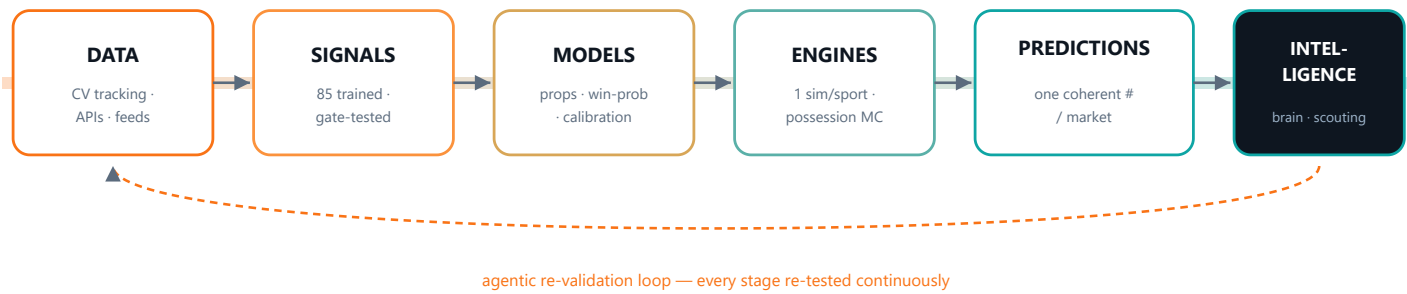
The product is a **calibrated predictor**, not a betting-edge product. The defensible win is matching the devigged closing line within noise on team-strength markets, plus a measured, calibrated *in-game conditioning* improvement. Beating the vig is not the goal; beating predictions — accuracy and calibration — is. An honest REJECT is a success, not a failure. No dollar edge is claimed anywhere.

What makes it unusual

- **It owns its data tier.** A real CV pipeline (YOLO → homography → tracking → re-ID → events) produces possession-level micro-state most public models never see.
- **It is honest by construction.** Hard-coded rails forbid fabricated edges, retracted numbers, dollar fields on the served product, and flag flips — even when running unattended for days.
- **It improves itself — carefully.** A 5-gate ratchet can recalibrate the served model, but only on changes that survive walk-forward, multi-corpus, clustered significance testing; otherwise it ships nothing and says so.
- **It was built by an agentic pipeline** directed by one human — Opus orchestrates and gates, a Sonnet fleet writes code, search agents explore — across ~1,470 commits.

2 • The Funnel

The spine of the whole system. Each stage narrows raw breadth into one validated, coherent number — and a loop re-tests every stage so nothing rots.



The funnel: breadth → validated narrowness → one coherent prediction → intelligence, wrapped in a self-re-testing loop.

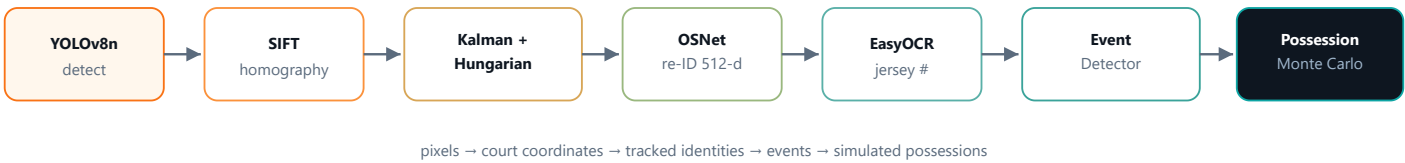
Stage	What happens	Discipline applied
Data	CV tracking output + keyless sports APIs + odds feeds + historical corpora.	Provenance tracked; never fabricated; offseason/empty states are honest, not faked.
Signals	~85 candidate predictive features proposed and trained; deep-data layers ingested per sport.	Each runs the leak-free gate; most REJECT and are kept as <i>scouting</i> , never force-fed.
Models	Player-prop models, win-probability, calibration layers (Platt / isotonic variants).	Walk-forward out-of-sample; held-out residuals drive uncertainty.
Engines	One simulation/rating engine per sport produces a single probability distribution.	Every served market is read off that one distribution (coherence asserted in code).
Predictions	One coherent number per market, devigged via Shin, with phase-labelled provenance.	<code>vs_close</code> labelled UNPROVEN; uncertainty surfaced; no dollar fields.
Intelligence	Obsidian brain + 80-artifact intelligence manifest + reject ledger + scouting.	Person-free, fully cross-linked, self-policing graph gates.

Why the funnel is the hard, original part

Anyone can fit a model. The originality here is the *end-to-end discipline*: making every data point earn its place through the same brutal gate, fusing only survivors into one coherent number per market, and keeping the whole thing honest while it runs and improves itself. That full sports-detail funnel — CV tier included — is what had to be invented from scratch.

3 • The Computer-Vision Foundation

The data tier most forecasters never own. From broadcast or tracking video, CourtVision reconstructs possession-level micro-state — the raw material of genuinely new signal.



The CV pipeline: a real, headless, GPU-bound stack turning video into possession-level state.

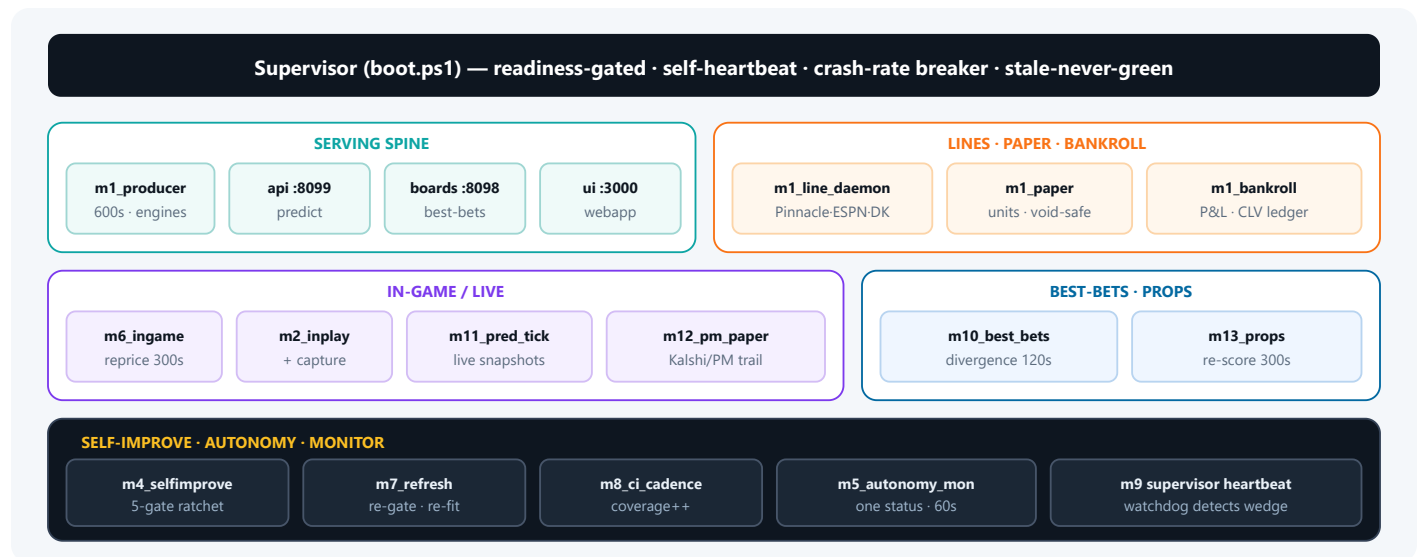
Stage	Component	Role
Detection	YOLOv8n	Players + ball per frame.
Geometry	SIFT homography (panorama)	Map image pixels to real court coordinates.
Tracking	Kalman filter + Hungarian assignment	Stable identities frame-to-frame.
Re-identification	OSNet (512-dim) + team-color tracker	Recover identity through occlusion / camera cuts.
Numbers	EasyOCR	Read jersey numbers (the current accuracy wall).
Semantics	EventDetector	Shots, possessions, transitions.
Simulation	Possession Monte Carlo	Roll possessions forward into win/score distributions.

The honest limit (and why it is the opportunity)

Jersey-OCR identity is noisy (~a low single-digit error wall), so the per-player CV micro-state is currently *identity-walled*: team and possession aggregates are clean, but clean per-player identity at scale is a real CV research problem, not a download. Solving it is exactly where the moat deepens — see §11.

4 · System Architecture

One command brings up a self-healing stack of 18 supervised daemons. Each has a heartbeat and an independent restart policy: if one dies it goes red alone and auto-restarts in seconds while the rest keep running.



The 18-daemon supervised stack. Independent branches: one dead daemon = one red entry, auto-restarted; the rest run on.

Operational properties

- **Independent & self-healing.** Kill any daemon → it returns in ~30s; kill the whole thing → clean stop leaves zero processes.
- **Stale-never-green.** A hung or dead daemon reads RED, never falsely "fresh." Heartbeat freshness windows are tuned per daemon cadence.
- **Boot-on-login.** A registered Windows task brings the whole stack up at logon via a watchdog that keeps the supervisor alive; governance preflight runs first; real money stays default-DENY.
- **Atomic writes** on every shared artifact so a crash mid-write never corrupts the served state.

5 • The Prediction Engines

Each sport has exactly one engine producing one probability distribution. Every market — moneyline, totals, spread, sets, props — is read off that same distribution, so the served numbers are internally coherent by construction (and the code asserts it).

Sport	Engine	One distribution → many markets
NBA	Elo + possession-level Monte Carlo	Win prob, totals, spread off one possession sim.
MLB	Margin-of-victory Elo + anchored Negative-Binomial	Run-line at 0 = moneyline; totals partition asserted.
Soccer	Exponentially-weighted Poisson / Dixon-Coles	All markets off one Dixon-Coles score matrix.
Tennis	Surface-Elo + point-level Monte Carlo	9 coherent markets off one MC matrix (sets, handicaps, scores).

Coherence is enforced, not hoped for

The producer asserts `served_prob == engine_prob`. A corrupted engine cell fails the coherence check by design — you cannot ship a market that disagrees with the distribution it came from.

Devigging & uncertainty

Market prices are devigged via **Shin**. Uncertainty bands come from held-out residuals — not invented. `vs_close` is labelled **UNPROVEN** everywhere: the system claims it *matches* the close, never that it beats it.

Above the engines sit player-prop models and calibration layers. Props synthesize from on-disk gamelogs when an external line feed rate-limits, so the board is never empty; every card carries `edge_claimed=false` and shows units, never dollars.

6 • In-Game Intelligence

This is the one place with a *validated* measured advantage — and the genuine frontier for everything beyond matching the close.

The proven part: prior-conditioning calibration

The served in-game number blends the **proven pregame prior** with realized game state (margin/run-diff, fraction elapsed). Conditioning the static prediction on live state produces a real, measured calibration improvement — statistically decisive on NBA fine-resolution data (Diebold–Mariano $p \approx 1.2 \times 10^{-12}$), holding across 3 of 4 sports. This is calibration, labelled as such — not a dollar claim.

Engineering note — carrying the prior across provider-id gaps

The live feed and the pregame snapshot sometimes key the same game under different provider id namespaces, which silently dropped the proven prior (degrading to a weaker base model). A leak-free resolver now bridges this by a *unique team match* (`PRIOR_TEAMS`) — exact id first, and on a tie (e.g. a doubleheader) it returns `BASE` rather than `guess`. Result: the served in-game number carries the proven prior for more games. Verified on real data; necessary input for the CLV frontier below.

The unproven frontier: in-game CLV

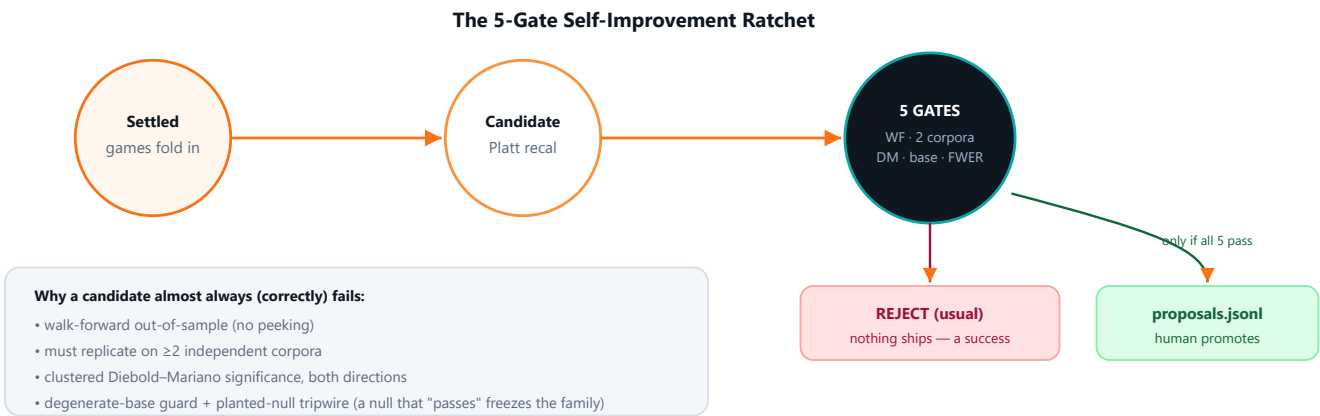
The open question — the real upside — is whether that calibration advantage converts into positive **closing-line value** against real liquid in-play prices (keyless Kalshi per-game markets, liquidity-gated). The capture loop pairs the model number with the devigged price every tick across live games; an aggregate grader pools per-game CLV with a game-resampling bootstrap and clustered Diebold–Mariano.

The bar for calling it an edge

BEAT requires ≥ 5 settled games, ≥ 40 ticks, a settled-outcome arm, a positive pooled-CLV confidence-interval lower bound, *and* cross-corpus both-direction significance. A pure market-copy candidate is structurally impossible to pass (it yields no corpus). Until that bar is cleared and replicated, `vs_close` stays UNPROVEN and real money stays DENY. Today: single-corpus, tiny-n — **INSUFFICIENT** .

7 · How It Gets Better

"Better" is three distinct things with different levers. Conflating them is how fake edges get born; separating them is how this system stays honest.



Sandbox-proven: on a planted miscalibration the ratchet cuts sealed-fold Brier 0.2025 → 0.0958 (seed-stable). On the real efficient market it returns NO_CANDIDATE — the honest, correct result.

Kind of "better"	What changes	Lever / gate	Status
1 · System	Validation coverage, robustness, the loop itself	Automatic, continuous (m8 cadence)	RUNNING
2 · Model	Served calibration actually changes	Data volume + 5-gate pass + human promote	READY / data-gated
3 · Edge	CLV proves positive vs real prices	Many in-season games + replication	UNPROVEN

The three improvement engines

- **Living refresh (hourly):** newly-settled finals fold into each sport's corpora; the in-game model is re-gated and re-fit, and either swaps in or downgrades its own provenance label. Continuous recalibration on fresh truth.
- **The 5-gate ratchet (daily):** the only path that can change a served number — and it never auto-ships; survivors land in `proposals.jsonl` for a human to promote.
- **CLV as a second corpus:** settled closing-line value is leak-free and gate-able, fed to the ratchet as a do-no-harm second corpus — never a ship objective (that is the market-follow trap, structurally blocked).

What "gets better" honestly means

Growing validation coverage and a recalibration engine *proven ready* that improves calibration when the data earns it — **not** a fake live-accuracy or edge climb. The served model is static until a gate-passing candidate is promoted. The expected daily outcome is REJECT, and that is the system working: an honest null beats a fabricated edge.

8 · The Validation Discipline & Honesty Rails

The credibility of every claim rests on a gate that rejects almost everything. The discipline is the product.

The gate, in full

- **Leak-free:** features are as-of, the outcome is never an input, the close is held out; verified on disk (zero debut-row leaks).
- **Walk-forward / out-of-sample:** single-fold lifts are treated as artifacts until they replicate.
- **≥ 2 independent corpora, both directions:** a signal must survive $A \rightarrow B$ and $B \rightarrow A$, not just one slice.
- **Clustered Diebold–Mariano by game:** significance that respects within-game correlation.
- **FWER budget + planted-null tripwire:** the bar tightens as more candidates are tested; a null that "passes" freezes its whole family — proof the gate has teeth.
- **Degenerate-base guard & collinearity check:** no winning by a broken baseline or by restating the base.

Numbers that are permanently retracted

Certain early figures were measurement artifacts (market-follow grading, a Q4 leak, an L5-proxy ceiling) and are **never** presented as current results. They appear only inside an explicit retraction context. Honesty is enforced, not aspirational: the served product carries no dollar fields, and a single boolean flips the whole status to "not honest" if a dollar/edge/stale-green/real-money leak ever appears.

The binding invariants (held even unattended, for days)

Never claim a dollar edge or print a retracted number.

Never auto-ship a model change or flip a feature flag on.

Never enable real money — it stays default-DENY.

Never push private data/ledgers to the public origin.

Always stale-never-green: dead reads red.

Always record honest rejects — they are the deliverable.

9 · The Intelligence Layer & Memory

Beneath the predictions sits a navigable knowledge graph — the system's long-term understanding of the sport — plus an auto-memory that lets the agentic builder carry lessons across sessions.

The Obsidian Brain

- **872 notes**, 100% reachable, one connected component, person-free.
- Player dossiers, scheme atlases, archetype hubs, comparables, game plans, leaderboards, lineup chemistry.
- An 80-artifact intelligence manifest + a reject ledger (the signal graveyard).
- Three self-policing graph gates keep it coherent on every rebuild.

Auto-Memory (cross-session)

- One fact per file: user profile, feedback, project state, references.
- Lets the agentic pipeline remember decisions, landmines, and "do-not-claim" rules between runs.
- Verified against the live code before reuse — memory informs, it never overrides what the repo actually says.

Scouting vs serving — the key distinction

Most ingested signals REJECT at the gate. Rejected signals are not discarded — they become **validated scouting**: recorded, searchable intelligence about what does and does not move a sport. They are *never force-fed* into the served prediction, because force-feeding a rejected signal is exactly the overfitting / market-follow failure mode the whole discipline exists to prevent. The intelligence layer is therefore an honest map of the sport's predictability frontier, not a pile of unused features.

The meta-finding (the plateau)

Tested at the finest available on-disk grids across all four sports — coarse state → at-bat / per-pitch / Statcast → CV aggregates → playstyle correlations — deeper data **rejects** for held-out win-probability calibration beyond the (margin/time) base plus the pregame prior. Recording that plateau, under gates proven to both fail noise and pass real signal, *is* a genuine scientific deliverable. One confirmed survivor (a soccer corners signal) cleared the hardest six-control gate — tiny in magnitude, real in fact, and still labelled calibration-only.

10 · Honest Current State

What is proven, what is unproven, and what is simply running — stated plainly.

Area	Status	Honest reading
Multi-sport parity	GREEN	All 4 sports + intl pass the census/manifest/feature-spec checks.
Pregame calibration	MATCHES CLOSE	Matches the devigged close within noise. No edge — and none claimed.
In-game conditioning	VALIDATED	Real measured calibration lift (DM $p \approx 1.2e-12$ NBA). Calibration, not dollars.
In-game CLV / "edge"	UNPROVEN	+1.0% mean CLV on n=13 settled — tiny, single-corpus, within noise.
Self-improve loop	ARMED · INERT	Runs the ratchet daily; returns REJECT/NO_CANDIDATE; ships nothing.
Paper bankroll	UNITS ONLY	An honest paper track record in units; <code>executed=false</code> ; not profit.
Real money	DENY	Stays off until CLV is replicated-positive. By design.

Why so much "unproven" is a feature

A system that mostly rejects, mostly says "insufficient," and refuses to claim a dollar edge is a system you can trust the rare time it says "this survived." The honesty is not timidity — it is what makes the eventual signal worth something. The market is efficient; matching it cleanly, and measuring an in-game advantage carefully, is the real and defensible achievement.

11 · Where It Can Go

This is the start, not the finish. The foundation is built and honest; the frontier is wide. Below: the founder's thesis, with proven foundation and aspirational frontier clearly separated.

The thesis

An originated, end-to-end sports-detail funnel — CV tracking through one coherent prediction through a self-validating loop — is a genuinely new kind of system. It is hard precisely because it has not been done this way before. The asset compounds: every day of in-season data, every deeper data tier, and every improvement to the CV identity layer feeds the *same brutal gate*, and the survivors are real because the gate is real.

The roadmap

Horizon	Move	Honest framing
Now → in-season	Volume on the in-game CLV loop; let the aggregate grader reach a real BEAT/MATCH/BEHIND verdict.	The immediate frontier. If CLV replicates positive → a real, measured edge. If not → a rigorous null. Either is a win.
Months	Deepen the data tier: Tier-3 acquisition (point-by-point, vendor tracking/xG, full Statcast) feeding the same gate.	Honest base rate: most still reject. The point is to find the few that don't — and capital funds this.
The moat	Integrate CV tracking deeply: solve the jersey-identity wall → clean per-player possession micro-state no public model owns.	This is the original differentiator. Person-level CV signal is the data nobody else can download — it must be built.
Platform	The sport-blind kernel + per-sport adapters generalize beyond the first four sports.	From "NBA tool" to a domain-agnostic forecasting + decision engine.
The builder	The agentic pipeline itself — discover → validate → ship/reject — keeps compounding the system unattended.	A system that gets more complete and better-validated on its own, within the rails.

The honest ceiling — and why it is a good one

Pregame: match the close (essentially reached — a real, defensible result). In-game: a *small* measured conditioning edge whose conversion to CLV is the open prize; if it replicates in liquid markets, that is rare and valuable precisely because almost everything else rejects. As intelligence and product: a high and largely realized ceiling — the most-complete, honestly-calibrated, self-validating multi-sport forecaster, with a CV data tier as its moat. The path to "edges on its own" is in-season volume + deeper data + the CV layer, all poured into the same uncompromising gate. Capital accelerates the data and the tracking; it does not change the discipline — and the discipline is what makes it true.

What to do next, concretely

1. Leave it running through in-season volume — the CLV verdict needs games, not code.
2. Acquire deeper data tiers and feed them to the existing gate (expect mostly rejects).
3. Invest in the CV identity layer — the one data source nobody can buy.

4. Promote only gate-survivors; keep real money DENY until CLV is replicated-positive.
5. Generalize the kernel/adaptor platform to new sports.

12 · Appendix

Glossary

Term	Meaning
CLV	Closing-line value — did you get a better number than the closing price. The honest track-record yardstick.
Devig / Shin	Removing bookmaker margin from prices to recover an implied probability; Shin is the method used.
Diebold–Mariano (DM)	A statistical test for whether one forecast is significantly more accurate than another.
Brier / BSS	Brier score measures probabilistic calibration (lower is better); BSS is its skill score vs a baseline.
Walk-forward	Train on the past, test strictly on the future — no peeking.
FWER	Family-wise error rate control — tightening the bar as more candidates are tested.
Planted-null	A known-fake signal injected to confirm the gate rejects noise (a tripwire).
INERT	The self-improve loop runs but ships nothing — proposal-only until a human promotes.
p0 / PRIOR_TEAMS	The leak-free pregame prior carried into the in-game number; PRIOR_TEAMS = resolved by a unique team match across provider-id gaps.
Scouting	A gate-rejected signal kept as validated intelligence, never force-fed into serving.

The stack at a glance

YOLOv8n → SIFT homography → Kalman+Hungarian → OSNet re-ID → EasyOCR → EventDetector → possession Monte Carlo → 85 signals → per-sport engine → FastAPI (predict :8099 · boards :8098) → webapp :3000 → 18-daemon supervisor → 5-gate self-improve ratchet → Obsidian brain + auto-memory.

The one-line truth

CourtVision is a self-healing, self-validating, honestly-calibrated multi-sport prediction engine with its own computer-vision data tier. It matches the close, measures a real in-game calibration advantage, improves itself only on what survives a brutal gate, and never claims an edge it has not proven. The edge frontier is open; the foundation is real; this is the start.

CourtVision System Bible · architect: Neel Shah · honest-by-construction edition · calibration not profit · no dollar edge claimed · real money default-DENY.